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Executive summary of study results: ☐ See Attached ☐ N/A

The purpose of this study was to test the ability of the mouse anti-lysophosphatidic acid (LPA) monoclonal antibody (mAb) to attenuate pain behavior in a mouse model of chemotherapy-induced peripheral neuropathy. Following baseline measurements, male DBA/2 mice were injected intraperitoneally (i.p.) every other day with paclitaxel or vehicle for a total of 4 doses. The anti-LPA mAb B3 or control IgG (30 mg/kg) was administered subcutaneously (s.c.) on day 3 as a post-treatment prior to the second dose of paclitaxel (therapeutic mode). On day 21 after 5 doses had been administered, a crossover study was initiated in which the group receiving control IgG was switched to B3 while the group receiving B3 was switched to IgG for an additional 4 doses. The development of tactile allodynia was assessed in the paw as the primary indicator of efficacy at defined intervals prior to and following paclitaxel and antibody dosing in all groups. An additional group of paclitaxel-treated mice received Gabapentin (100 mg/kg i.p.) or PBS during the early phase on day 3 and the late phase on day 20 as a reference standard for the study.
Results showed that B3 significantly attenuated the paclitaxel-induced reduction in paw withdrawal threshold compared with both IgG control and PBS post-treatment. Gabapentin also reduced paclitaxel-mediated allodynia.
In summary, this study demonstrated efficacy of the murine anti-LPA antibody to reverse paclitaxel-induced tactile allodynia when administered either during or following completion of chemotherapeutic treatment.

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Efficacy of murine anti-LPA mAb B3 in a mouse model of paclitaxel-induced peripheral neuropathy

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1 SUMMARY

The purpose of this study was to test the ability of the mouse anti-lysophosphatidic acid (LPA) monoclonal antibody (mAb) to attenuate pain behavior in a mouse model of chemotherapy-induced peripheral neuropathy. Following baseline measurements, male DBA/2 mice were injected intraperitoneally (i.p.) every other day with paclitaxel or vehicle prepared according to the method of clinical formulation for a total of 4 doses. The anti-LPA mAb B3 or control IgG (30 mg/kg) was administered subcutaneously (s.c.) on day 3 as a post-treatment prior to the second dose of paclitaxel (therapeutic mode). On day 21 after 5 doses had been administered, a crossover study was initiated in which the group receiving control IgG was switched to B3 while the group receiving B3 was switched to IgG for an additional 4 doses. The development of tactile allodynia was assessed in the paw as the primary indicator of efficacy at defined intervals prior to and following paclitaxel and antibody dosing in all groups. An additional group of paclitaxel-treated mice received Gabapentin (100 mg/kg i.p.) or PBS during the early phase on day 3 and the late phase on day 20 as a reference standard for the study.

Results showed that B3 significantly attenuated the paclitaxel-induced reduction in paw withdrawal threshold compared with both IgG control and PBS post-treatment. Gabapentin also reduced paclitaxel-mediated allodynia, yet appeared to exhibit greater apparent potency in the early phase. In summary, this study demonstrated efficacy of the murine anti-LPA antibody to reverse paclitaxel-induced tactile allodynia when administered either during or following completion of chemotherapeutic treatment.

2 INTRODUCTION

The bioactive lipid lysophosphatidic acid (LPA) is an important extracellular signaling molecule in neuropathic pain. LPA and its receptors are responsible for long-lasting mechanical allodynia and thermal hyperalgesia as well as demyelination and upregulation of pain-related protein expression patterns in a variety of animal models. Evidence supports the hypothesis that antagonism of the LPA signaling pathways would benefit the treatment of neuropathic pain. Pain is a common symptom in cancer patients and may be caused by both tumor-related pathologies as well as chemotherapeutic agents used for treatment. Paclitaxel is an effective anti-neoplastic agent that has been recognized to produce a dose-limiting painful peripheral neuropathy in a clinically significant number of cancer patients. Models of paclitaxel-induced neuropathic pain have been developed and both allodynia and thermal hyperalgesia can be detected in such models. We have developed the first monoclonal antibody targeting LPA (anti-LPA Ab). In preliminary studies, our anti-LPA mAb has demonstrated efficacy in attenuating pain behaviors in diabetic, chemotherapeutics, arthritis and sciatic nerve injury induced-neuropathic pain. Furthermore, therapy with anti-LPA Ab provides a long-lasting, sustained effect in accordance with the pharmacokinetic profile of the antibody.

3 MATERIALS AND METHODS

3.1 Test Article and Control

Murine anti-LPA antibody (B3) Lot# HR121338

Murine nonspecific mAb control (1014) Lot# LP91145

3.2 Study Design

After stable baseline von Frey threshold measurements were obtained over 2-3 consecutive days, allodynia was induced in male DBA/2 mice by i.p. injection of paclitaxel 1 mg/kg or vehicle (1:1:18 clinical formulation of 5% Cremaphor EL, 5% anhydrous ethanol in phosphate-buffered saline) on days 1, 3, 5 and 7 in a volume of 10 ml/kg for a cumulative paclitaxel dose of 4 mg/kg. The development of tactile allodynia was assessed as the primary indicator of efficacy at defined intervals after injection of paclitaxel (days 3, 5 and 7) and then twice weekly prior to antibody dosing in all groups for up to 5 weeks. The control groups received paclitaxel according to the schedule below and were treated once with acute gabapentin (positive control) or saline (negative control) in the early phase on day 3 and again in the late phase on day 20. Groups receiving antibody began treatment with nonspecific murine monoclonal antibody as a control (1014) or anti-LPA antibody B3 at 30 mg/kg s.c. prior to the second dose of paclitaxel on day 3, followed by dosing at a frequency of twice per week for 5 doses. To examine if B3 demonstrates efficacy as a post-treatment during the late phase and to determine if the effects of early dosing with B3 persisted during the washout period, the 1014 group was switched to B3 and vice-versa at a frequency of twice per week for 4 doses. Body weights were monitored 2-3 times per week throughout the study. At the conclusion of the study, all mice were sacrificed and blood and tissues were collected 3 days after the last dose of antibody to ensure that secondary end points and markers could be correlated with efficacy readouts.

Table 1. Study design

Table 1: Treatment Groups				
Group	No. Animals	Route	Dose (mg/kg)	Treatment Schedule
Vehicle (1:1:18 Cremaphor-EL: ethanol: PBS)	10	IP	1 ml/Kg	Days 1, 3, 5 and 7
paclitaxel/ Acute saline	10	IP IP	1 mg/Kg 100 ml/Kg	Days 1, 3, 5 and 7/ Days 3 (early) and 20 (late)
paclitaxel/ Acute gabapentin	8	IP IP	1 mg/Kg 100 mg/Kg	Days 1, 3, 5 and 7/ Days 3 (early) and 20 (late)
paclitaxel/ 1014 (early phase) or B3 (late phase)	10	IP SC	1 mg/Kg 30 mg/Kg	Days 1, 3, 5 and 7/ Day 3 onward @ 2x/week
paclitaxel/ B3 (early phase) or 1014 (late phase)	10	IP SC	1 mg/Kg 30 mg/Kg	Days 1, 3, 5 and 7/Day 3 onward @ 2x/week

3.3 Blood Collection and Analysis

Blood samples were collected by cardiocentesis at the termination of the study, placed in chilled tubes containing potassium EDTA (K_2EDTA) and kept on ice until processed. Samples were centrifuged at 5000 rpm for 10 minutes at 4ºC, and the supernatant containing plasma was harvested. Plasma samples were immediately frozen in dry ice and stored at -70 ± 10°C in aliquots until assayed. Anti-LPA antibody plasma concentrations were quantified using a direct-binding immunoassay (ELISA) method.

3.4 Tactile Response Threshold

For the assessment of tactile allodynia, DBA/2 mice were placed in individual Plexiglas compartments with a wire mesh floor and allowed to acclimate to the testing apparatus for 30-60 min prior to measurement. Measurements of tactile paw withdrawal thresholds (PWT) were conducted on both left and right hindpaws by an observer blinded to the treatment conditions prior to (at baseline) and following chemotherapeutic and/or drug delivery. Tactile PWT were measured with a series of von Frey filaments with buckling forces between 0.02 and 2 g (Stoelting, Wood Dale, IL) using the up-down method of Dixon, as described previously. Briefly, calibrated filaments beginning with a starting force of 0.4 g were applied to the mid-paw plantar surface until the filament was slightly bent (with a pressure corresponding to stimulation of the L4 dermatome) and maintained for up to 5 seconds. Lifting/flicking of the paw and/or spreading of digits were recorded as a positive response and the next lightest filament was chosen for the next measurement. Absence of a response after 5 seconds prompted use of the next filament of increasing force. Stimuli were separated by several seconds to prevent sensitization or until the animal was stationary with paws placed flat on the grid. This paradigm was continued until four measurements were made after an initial change in the behavior or until five consecutive negative (given the score of 2 g) or four positive (score of 0.02 g) scores occurred. The resulting sequence of positive and negative scores was used to interpolate the 50% response threshold. Since the treatments are systemic and paclitaxel-induced neuropathy manifests bilaterally, mean thresholds were calculated for left and right hindpaws of each animal and then averaged among each group. Mice with mean baseline (pre-injury) thresholds below 0.75 g were removed from the study. For the paclitaxel-treated groups, only mice with a 50% tactile response threshold below 0.5 g were considered allodynic and brought forward for drug testing.

3.5 Direct Binding ELISA

B3 antibody plasma concentrations were quantified using a direct-binding immunoassay (ELISA) method. 384-well plates were coated with 15 µl of 1 µg/ml (C12)LPA-BSA-SMCC for 1 hour at 37 degrees. The plates were washed 4 times with PBS and blocked with 50 µl of PBS-BSA-tween for 1 hour at room temperature. 15 µl of B3 antibody was added to each well beginning at a concentration of 4 µg/ml and diluted 2-fold across each plate for the standard curve. Mouse plasma samples were diluted to 1:100, 1:1000, 1:2000, 1:4000, 1:8000, and 1:16000 and 15 µl of each dilution (run in duplicate) was added to the plate and incubated at room temperature for 1 hour. Next, plates were washed 4 times with PBS and 15 µl of a peroxidase conjugated anti-mouse IgG (H+L) (Jackson immunoresearch 115-035-062) was added at a 1:40000 dilution and incubated for 1 hour at room temperature. The plates were washed 4 times with PBS and developed with 100 µl cold TMB substrate for 12 minutes at room temperature. Finally, the reaction was stopped with 15 µl of 1M H_2SO_4 and then the plates were read at 450 nm.

3.6 Detection of Bioactive LPA

Bioactive LPA was detected using The CHO-K1 EDG2 (LPA_1 receptor) β-arrestin clone (PathHunter™) that uses Enzyme Fragment Complementation technology in which two weakly complementing fragments of the B-galactosidase (β-gal) enzyme are expressed in stably transfected cells. One fragment of the β-gal, termed the enzyme acceptor (EA), is fused to the C-terminus of β-arrestin2. The complementing fragment of β-gal, termed the ProLink™ tag, is expressed as a C-terminal fusion protein with the LPA_1 receptor. Upon activation, the LPA_1 receptor is phosphorylated, providing a binding site for β-arrestin. The interaction of β-arrestin and the LPA_1 receptor forces the interaction of Prolink and EA, thus allowing complementation of the two fragments of β-gal and the formation of functional enzyme capable of hydrolyzing substrate and generating a chemiluminescent signal. Briefly, CHO-K1 EDG2 cells were plated at and incubated at 37°C/5% CO_2. After 24 hours, plates were starved with reduced serum media at 37°C for another 24 hours. LPA standard curves were prepared by titrating LPA (0 – 60µM) in F12 media containing 1mg/mL BSA onto the plate to determine low and maximal signal (dotted lines on Figure 2). Standards and samples were added to the plate and incubated at room temperature for 90 minutes. The plates were washed, incubated with the detection reagent in the dark for 90 minutes and read on a standard luminescence plate reader.

3.7 Data Analysis

All data was generated by Lpath and collected at BioTox Sciences and Lpath. Descriptive statistics was used to calculate median, mean, standard deviation (SD) and standard error of the mean (SEM) using GraphPad Prism software (version 6.0, Graph Software Inc, La Jolla, CA). GraphPad Prism software was used to graph the data. Ordinary one-way ANOVA was used for biochemical assays and body weight measures while two-way repeated measures ANOVA was used for behavioral analysis. Both tests were followed *post-hoc* by Bonferroni correction.

4 RESULTS

4.1 Body Weight

There was no significant difference in body weight (Table 2) between vehicle and paclitaxel-treated mice receiving either PBS or Gabapentin i.p. on days 3 and 20 or twice weekly 1014 or B3 30 mg/kg s.c. according to the schedule outlined in Table 1.

4.2 Mechanical Allodynia

Mechanical allodynia was fully developed in paclitaxel-treated versus vehicle-treated mice on day 3, or two days following the first dose of chemotherapeutic (see Figure 1). Once allodynia was confirmed, paclitaxel-treated mice were randomized into treatment groups and antibody dosing was begun as described in the study design and in Table 1. Results showed that s.c. delivery of B3 30 mg/kg twice weekly following the first dose of paclitaxel produced a significant ($p < 0.001$), sustained reversal of tactile allodynia in the early phase. In the crossover study, switching this group to s.c. IgG 30 mg/kg twice weekly prompted the return of allodynia after two doses. More importantly, B3 also reversed tactile allodynia ($p < 0.05$) when administered in the latter phase of the study, starting twice weekly at day 21 (14 days following completion of chemotherapy). The reference substance gabapentin (100 mg/kg i.p.) significantly ($p < 0001$) attenuated tactile allodynia both in the early and late phase of the study, although its apparent potency was greater on day 3 than on day 20 (Figure 2).

4.3 Bioactive LPA

Levels of bioactive LPA (defined as the amount of LPA present in plasma able to activate the LPA_1 receptor in the DiscoveRX reporter cell assay) were measured in the plasma of mice from groups in Table 1. Compared with vehicle, paclitaxel-treated mice displayed no change in LPA levels. However, B3 treatment significantly reduced levels of bioactive LPA in plasma (Figure 3).

Table 2 Body weights

VEHICLE

DAY	0	3	5	7	10	14	17	21	24	28	31	35
1-1	26.0	27.0	26.0	26.0	26.0	26.0	27.0	28.0	27.0	28.0	28.0	27.0
1-2	28.0	30.0	28.0	28.0	28.0	28.0	27.0	28.0	28.0	27.0	27.0	27.0
1-3	23.0	24.0	22.0	22.0	22.0	21.0	22.0	23.0	23.0	22.0	22.0	22.0
1-4	22.0	23.0	23.0	22.0	21.0	21.0	23.0	24.0	22.0	21.0	23.0	23.0
7-1	26.0	28.0	28.0	28.0	28.0	29.0	29.0	30.0	29.0	29.0	30.0	31.0
7-2	22.0	25.0	24.0	25.0	24.0	25.0	25.0	26.0	24.0	24.0	25.0	26.0
7-3	20.0	22.0	21.0	21.0	22.0	22.0	22.0	23.0	22.0	22.0	23.0	22.0
7-4	24.0	26.0	25.0	26.0	26.0	25.0	27.0	28.0	27.0	27.0	28.0	28.0
12-1	23.0	25.0	24.0	24.0	25.0	24.0	24.0	26.0	27.0	26.0	27.0	27.0
12-4	26.0	28.0	27.0	27.0	27.0	27.0	28.0	30.0	30.0	31.0	30.0	31.0
mean	24.0	25.8	24.8	24.9	24.9	24.8	25.4	26.6	25.9	25.7	26.3	26.4
sem	0.8	0.8	0.8	0.8	0.8	0.9	0.8	0.8	0.9	1.1	0.9	1.0

PACLITAXEL+PBS

DAY	0	3	5	7	10	14	17	20	24	28	31	35
2-1	29.0	30.0	29.0	28.0	29.0	28.0	29.0	30.0	30.0	31.0	32.0	32.0
2-2	26.0	27.0	27.0	26.0	26.0	26.0	27.0	27.0	28.0	28.0	29.0	30.0
2-3	24.0	25.0	24.0	23.0	23.0	23.0	24.0	24.0	25.0	26.0	26.0	25.0
2-4	29.0	30.0	29.0	28.0	28.0	27.0	29.0	30.0	30.0	31.0	32.0	32.0
6-1	24.0	25.0	25.0	24.0	25.0	26.0	27.0	26.0	28.0	28.0	27.0	28.0
6-2	22.0	23.0	23.0	23.0	24.0	24.0	25.0	24.0	25.0	26.0	25.0	26.0
6-3	22.0	23.0	23.0	23.0	24.0	24.0	25.0	24.0	25.0	25.0	25.0	25.0
6-4	21.0	23.0	22.0	22.0	23.0	23.0	23.0	22.0	24.0	23.0	23.0	24.0
mean	24.6	25.8	25.3	24.6	25.3	25.1	26.1	25.9	26.9	27.3	27.4	27.8
sem	1.1	1.0	1.0	0.8	0.8	0.7	0.8	1.0	0.9	1.0	1.2	1.1

Table 2 Body weights (continued)

PACLITAXEL+Gabapentin

DAY	0	3	5	7	10	14	17	20	24	28	31	35
3-1	24.0	25.0	23.0	23.0	23.0	22.0	23.0	23.0	23.0	23.0	24.0	24.0
3-2	25.0	26.0	25.0	25.0	25.0	26.0	26.0	26.0	27.0	27.0	28.0	28.0
3-3	27.0	28.0	28.0	26.0	26.0	26.0	27.0	27.0	28.0	29.0	29.0	29.0
3-4	26.0	28.0	26.0	26.0	27.0	27.0	27.0	28.0	28.0	29.0	30.0	31.0
13-1	27.0	28.0	28.0	28.0	28.0	29.0	29.0	28.0	30.0	30.0	30.0	31.0
13-2	30.0	30.0	29.0	30.0	31.0	32.0	32.0	32.0	34.0	36.0	35.0	35.0
13-3	24.0	26.0	26.0	26.0	26.0	27.0	27.0	28.0	29.0	29.0	29.0	30.0
13-4	24.0	25.0	24.0	24.0	25.0	25.0	26.0	25.0	26.0	26.0	27.0	27.0
mean	25.9	27.0	26.1	26.0	26.4	26.8	27.1	27.1	28.1	28.6	29.0	29.4
sem	0.7	0.6	0.7	0.8	0.8	1.0	0.9	0.9	1.1	1.3	1.1	1.1

PACLITAXEL+1014 switch to B3

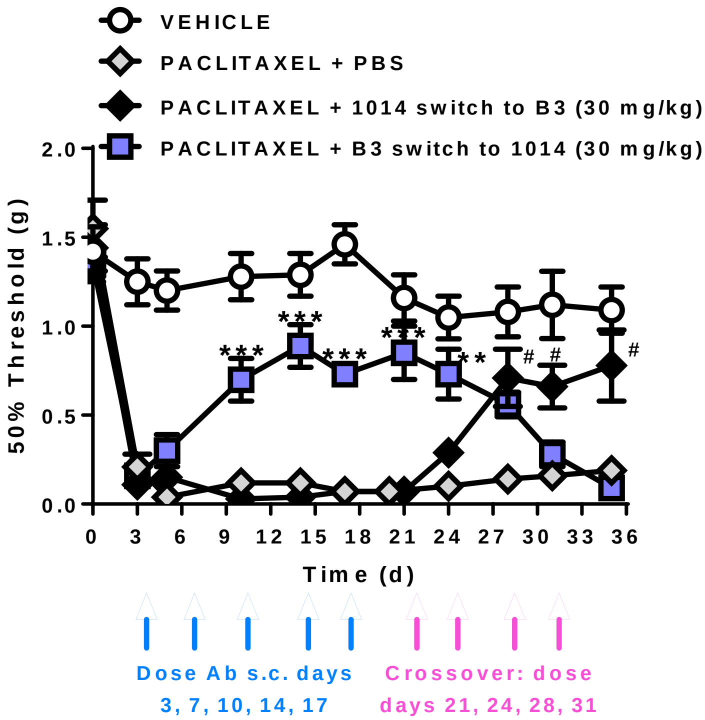
DAY	0	3	5	7	10	14	17	21	24	28	31	35
4-1	30.0	31.0	30.0	30.0	30.0	31.0	31.0	32.0	31.0	32.0	32.0	32.0
4-2	25.0	26.0	26.0	25.0	25.0	25.0	26.0	28.0	27.0	27.0	27.0	27.0
4-3	29.0	30.0	29.0	28.0	29.0	29.0	29.0	30.0	29.0	30.0	31.0	32.0
4-4	28.0	30.0	29.0	28.0	29.0	29.0	28.0	31.0	30.0	31.0	31.0	31.0
8-1	23.0	25.0	24.0	25.0	25.0	25.0	25.0	27.0	28.0	27.0	28.0	28.0
8-2	21.0	22.0	22.0	22.0	23.0	22.0	22.0	24.0	24.0	24.0	23.0	24.0
8-4	20.0	22.0	21.0	21.0	21.0	22.0	21.0	22.0	23.0	23.0	23.0	23.0
11-1	22.0	24.0	23.0	24.0	24.0	25.0	25.0	25.0	26.0	26.0	27.0	27.0
11-2	23.0	24.0	24.0	24.0	24.0	24.0	25.0	25.0	25.0	25.0	26.0	25.0
11-3	22.0	24.0	24.0	24.0	24.0	25.0	26.0	25.0	27.0	27.0	27.0	27.0
11-4	24.0	25.0	25.0	26.0	26.0	26.0	26.0	27.0	28.0	28.0	28.0	28.0
mean	24.3	25.7	25.2	25.2	25.5	25.7	25.8	26.9	27.1	27.3	27.5	27.6
sem	1.0	1.0	0.9	0.8	0.8	0.9	0.9	0.9	0.7	0.9	0.9	0.9

Table 2 Body weights (continued)

PACLITAXEL+B3 switch to 1014

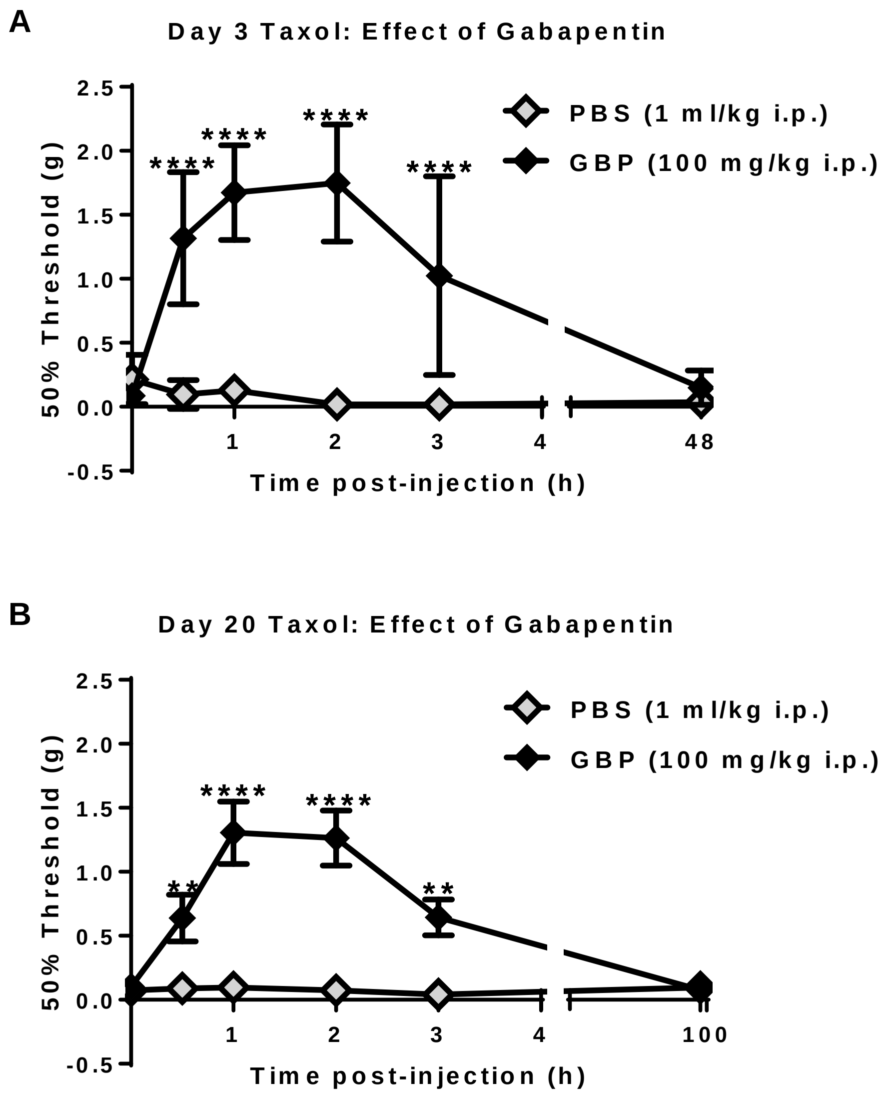
DAY	0	3	5	7	10	14	17	21	24	28	31	35
5-1	25.0	26.0	26.0	26.0	26.0	25.0	26.0	26.0	27.0	28.0	27.0	29.0
5-2	22.0	24.0	24.0	24.0	24.0	23.0	22.0	24.0	24.0	25.0	25.0	26.0
5-4	28.0	28.0	29.0	29.0	29.0	29.0	30.0	29.0	30.0	31.0	30.0	32.0
9-1	24.0	26.0	26.0	25.0	25.0	27.0	27.0	27.0	27.0	27.0	28.0	28.0
9-2	27.0	27.0	27.0	26.0	27.0	29.0	29.0	30.0	29.0	30.0	31.0	31.0
9-3	21.0	22.0	21.0	21.0	21.0	21.0	22.0	23.0	22.0	22.0	22.0	22.0
9-4	24.0	26.0	25.0	25.0	25.0	27.0	28.0	29.0	28.0	27.0	27.0	28.0
10-1	22.0	23.0	22.0	23.0	23.0	23.0	24.0	25.0	24.0	24.0	25.0	24.0
10-2	22.0	23.0	23.0	23.0	23.0	24.0	24.0	25.0	26.0	25.0	26.0	26.0
10-3	20.0	22.0	21.0	22.0	22.0	22.0	23.0	24.0	24.0	24.0	24.0	24.0
10-4	22.0	25.0	24.0	24.0	24.0	25.0	25.0	26.0	26.0	26.0	26.0	26.0
mean	23.4	24.7	24.4	24.4	24.5	25.0	25.5	26.2	26.1	26.3	26.5	26.9
sem	0.8	0.6	0.8	0.7	0.7	0.8	0.8	0.7	0.7	0.8	0.8	0.9

Figure 1 **Timecourse of paclitaxel-induced tactile allodynia in DBA/2 mice.**



Vehicle, 1:1:18 clinical formulation (5% Cremaphor EL, 5% anhydrous ethanol in PBS, phosphate-buffered saline). 1014, IgG control; B3, murine anti-LPA. # p <0.05, **p <0.01, ***p <0.001, v. paclitaxel + PBS by two-way repeated measures ANOVA with Bonferroni *post-hoc* analysis. Data are expressed as mean ± SEM, n =8-11/group.

Figure 2 Timecourse of gabapentin effect in paclitaxel-treated DBA/2 mice.



Timecourse of tactile thresholds measured at baseline and following intraperitoneal (i.p.) injection of PBS or GBP (Gabapentin) in paclitaxel-pretreated mice on **A)** Day 3 and on **B)** Day 20 of the study. $^{**}p < 0.01$, $^{****}p < 0.0001$ v. paclitaxel + PBS by two-way repeated measures ANOVA with Bonferroni *post-hoc* analysis. Data are expressed as mean ± SEM, n =8/group.

4.4 Antibody Concentration at Takedown

The antibody concentration in plasma of mice treated with B3 antibody or 1014 was as expected for the dose administered as measured by B3 direct ELISA (Table 3).

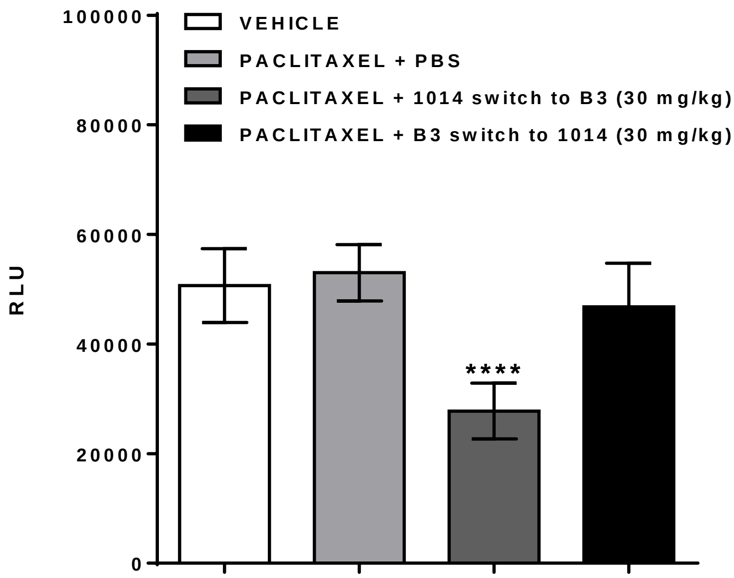
Table 3 Terminal concentrations of B3 in plasma

Paclitaxel + 1014 switch to B3 (30 mg/kg)	
mouse ID	[B3] (μg/mL)
13	410.92
14	299.25
15	204.78
16	279.78
29	166.30
30	270.75
32	357.57
41	306.22
42	128.45
43	237.85
44	446.15
Mean	282.55

Paclitaxel + B3 switch to 1014 (30 mg/kg)	
mouse ID	[B3] (μg/mL)
17	12.92
18	4.30*
20	16.81
33	10.39*
34	22.02
35	16.93
36	7.34*
37	4.98*
38	7.25*
39	8.53*
40	36.09
Mean	13.41

*Below standard linear range

Figure 3 Levels of bioactive LPA in DBA/2 mice following paclitaxel, 1014 or B3



LPA levels were measured by EDG2 assay from plasma samples obtained from mice treated as described above. ****p <0.0001 v. paclitaxel + PBS by one-way ANOVA with Bonferroni *post-hoc* analysis. Data are expressed as mean ± SEM, n =8-12/group.

5 CONCLUSION

Therapeutic intervention with the murine anti-LPA antibody B3 both decreased the circulating levels of bioactive LPA and attenuated tactile allodynia in a mouse model of paclitaxel-induced peripheral neuropathy.

6 DATA ARCHIVE

Lpath Notebooks #295, pp. 39-58 and #308, pp.68-69, 130-132; Binder #295, pp. 178-254.

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